

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

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THE PATENTS RULES, 2003

**COMPLETE
SPECIFICATION**

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(See section 10 and rule 13)

TITLE OF INVENTION

**MACHINE LEARNING-BASED BATTERY TEMPERATURE FAULT
DETECTION SYSTEM FOR ELECTRIC VEHICLES**

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THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES

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THE INVENTION AND THE MANNER IN WHICH IT IS TO BE

PERFORMED

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to battery management systems for electric vehicles, and more particularly to a machine learning-based system for
10 detecting abnormal battery temperature conditions using ensemble algorithms including Random Forest, XGBoost, LightGBM, and CatBoost to enhance safety and prevent thermal runaway in lithium-ion battery packs.

BACKGROUND

[0004] The following text of background art is provided for purposes of
15 understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Electric vehicles have gained widespread adoption as a sustainable transportation solution, with lithium-ion batteries serving as the primary energy storage technology due to their high energy density and long cycle life. Battery
20 Management Systems (BMS) monitor parameters such as voltage, current, temperature, and state of charge to ensure safe operation and prevent failures. Traditional BMS approaches rely on threshold-based monitoring and rule-based algorithms to detect battery anomalies, where predefined temperature limits trigger protective actions when exceeded.

25 [0006] Machine learning techniques have emerged as powerful tools for battery health monitoring and fault detection. Various algorithms including support vector machines, neural networks, and ensemble methods have been applied to analyze battery sensor data and identify patterns indicative of potential failures. Random Forest, XGBoost, LightGBM, and CatBoost represent advanced ensemble
30 learning approaches that combine multiple decision trees to improve prediction

accuracy and robustness. These methods have shown promise in processing large datasets and capturing complex relationships between battery parameters.

[0007] Temperature monitoring represents a fundamental aspect of battery safety, as thermal runaway can lead to fires, explosions, and system failures. Zhao et al. developed hybrid neural networks for online surface temperature prediction and abnormal diagnosis of lithium-ion batteries using fault threshold optimization. Chang investigated enhanced power battery fault detection in new energy vehicles using LightGBM algorithms. Li et al. applied Random Forest regression for online capacity estimation of lithium-ion batteries, while Jafari and Byun explored optimized XGBoost modeling for accurate battery capacity degradation prediction.

[0008] However, conventional threshold-based fault detection methods struggle to handle the nonlinear behavior and dynamic characteristics of lithium-ion batteries under varying operational conditions. These approaches often fail to detect subtle early-stage faults and may generate false alarms due to their inability to adapt to changing battery conditions. Rule-based systems lack the flexibility to account for complex interactions between multiple battery parameters and environmental factors that influence thermal behavior.

[0009] Current machine learning approaches for battery fault detection face limitations in real-time implementation and generalization across different battery chemistries and operating environments. Many existing models are trained on narrow datasets and may not perform reliably when applied to diverse real-world scenarios. The computational complexity of some algorithms poses challenges for deployment in embedded battery management systems with limited processing resources.

[0010] It has been appreciated that a system is needed that overcomes one or more of these problems.

[0011] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

SUMMARY

[0012] The present invention introduces a machine learning-based battery temperature fault detection system for electric vehicles. This system comprises temperature sensors to monitor lithium-ion battery packs, a data acquisition module to collect real-time sensor data, and a preprocessing module for data cleaning and normalization. A machine learning unit employs ensemble learning algorithms (Random Forest, XGBoost, LightGBM, CatBoost) to analyze data, a fault detection module identifies abnormal temperatures, and an alert system triggers warnings for unsafe conditions.

10 **[0013]** The use of multiple ensemble learning algorithms enhances fault detection accuracy by leveraging their unique strengths. Random Forest reduces overfitting, XGBoost corrects errors sequentially, LightGBM offers computational efficiency, and CatBoost handles categorical data effectively. This multi-algorithm approach outperforms single algorithms and conventional threshold-based systems.

15 **[0014]** The data preprocessing module applies outlier detection, interpolation for missing values, and standardization for feature scaling, ensuring high-quality input data. The machine learning unit trains algorithms using an 80-20 train-test split and evaluates performance with metrics like Mean Absolute Error and R-squared score. This method ensures robust model validation and unbiased performance assessment.

20 **[0015]** The fault detection module implements dynamic thresholding, adjusting sensitivity based on operational conditions such as load states and battery aging. This reduces false alarms and maintains high fault detection accuracy. Temperature sensors, strategically placed on the battery pack, provide comprehensive thermal monitoring, enabling real-time processing and immediate fault detection.

25 **[0016]** The alert generation system offers graduated alert levels for minor deviations and critical conditions. This enables preventive maintenance and immediate safety protocols. The ensemble learning algorithms achieve high classification accuracy and regression performance, ensuring reliable fault detection and minimal false positives or negatives.

[0017] The method involves collecting real-time sensor data, preprocessing it, training multiple ensemble learning algorithms, and evaluating their performance. The algorithm with the best performance is deployed for fault detection. Dynamic threshold detection adjusts sensitivity based on operational conditions, and real-time monitoring compares predicted and actual sensor readings, generating alerts for abnormal conditions.

[0018] A computer-readable storage medium stores instructions for collecting sensor data, preprocessing it, implementing ensemble learning algorithms, detecting abnormal temperatures, and generating alerts. This software-based approach allows flexible deployment, updates, and enhancements without hardware modifications, ensuring scalability and maintainability.

[0019] The system includes hyperparameter optimization for each algorithm to maximize fault detection performance. Adaptive learning capabilities continuously update model parameters based on new data, maintaining accuracy as battery characteristics change over time. This ensures the system remains effective against emerging fault modes and evolving battery technologies.

[0020] The described invention provides a robust and scalable solution for battery temperature fault detection in electric vehicles, leveraging advanced machine learning techniques and dynamic thresholding to ensure high accuracy, real-time monitoring, and effective alert generation.

BRIEF DESCRIPTION OF FIGURES

[0021] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

[0022] The illustrations are schematic in nature and serve solely to clarify the principles and functionality of the present disclosure. While particular constructions and configurations are depicted for explanatory purposes, it will be understood that the invention is not limited to the precise implementations shown in the figures and may encompass variations without departing from the scope of protection.

[0023] FIG. 1 illustrates a Random Forest ensemble learning methodology for battery fault detection, according to aspects of the present disclosure.

[0024] FIG. 2 depicts an ensemble machine learning architecture with multiple decision trees for battery fault prediction, according to an embodiment.

5 [0025] FIG. 3 illustrates a leaf-wise tree growth strategy for gradient boosting frameworks, according to aspects of the present disclosure.

[0026] FIG. 4 depicts a systematic fault detection and monitoring workflow for a battery management system, according to an embodiment.

[0027] FIG. 5 illustrates a comprehensive machine learning pipeline for battery
10 fault detection and analysis, according to aspects of the present disclosure.

[0028] FIG. 6 depicts a line graph showing battery temperature fault detection with difference analysis, according to an embodiment.

[0029] FIG. 7 illustrates a comparison of temperature behavior under normal and abnormal battery operating conditions, according to aspects of the present
15 disclosure.

[0030] FIG. 8 depicts a scatter plot comparing actual versus predicted temperature values under normal conditions, according to an embodiment.

[0031] FIG. 9 illustrates a scatter plot comparing actual versus predicted battery temperature values under abnormal conditions, according to aspects of the
20 present disclosure.

Furthermore, in terms of the construction of the system, one or more components of the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the figures with details that will be readily
25 apparent to those of ordinary skill in the art having benefit of the description herein.

DETAILED DESCRIPTION

[0032] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to
30 the accompanying drawings, the invention is not limited to the particular forms

disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

5 **[0033]** The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

10 **[0034]** Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

15 **[0035]** The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

20 **[0036]** The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures 1. In Figures 1, the same elements or elements that have the same functions are indicated by the same reference signs.

30 **[0037]** The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0038] The battery monitoring system employs a distributed thermal sensing architecture comprising three temperature sensors designated T1, T2, and T3. These sensors are positioned at strategic locations across the battery pack to provide comprehensive thermal distribution monitoring. The T1 sensor is positioned at a first location on the battery pack, the T2 sensor is positioned at a second location on the battery pack, and the T3 sensor is positioned at a third location on the battery pack. This multi-point sensing configuration enables detection of localized temperature variations and thermal gradients that may indicate developing fault conditions or uneven thermal distribution across the battery assembly.

[0039] The battery monitoring system utilizes DMEC INR18650-26E lithium-ion cells as the primary energy storage components. Each lithium-ion cell has a nominal voltage of 3.6V and a nominal capacity of 2.6Ah. The lithium-ion cells are rated for a discharge current of 15A, providing adequate power delivery for electric vehicle applications. The physical dimensions of each lithium-ion cell are 18.45mm in diameter and 65.2mm in length. Each lithium-ion cell has a weight of 47 grams, contributing to the overall energy density characteristics of the battery pack. The internal resistance of each lithium-ion cell ranges from 0.03m Ω to 0.05m Ω , which affects the thermal characteristics and power delivery efficiency during operation.

[0040] The experimental hardware setup comprises several components configured to simulate real-world electric vehicle battery usage conditions. A DC motor is connected to the battery pack to provide a controllable electrical load. A speed controller is operatively connected to the DC motor to regulate the motor speed and corresponding current draw from the battery pack. A loading rheostat is incorporated into the circuit to provide variable resistance and enable simulation of different load conditions. The loading rheostat allows adjustment of the current draw to replicate various driving scenarios and power demands that would be encountered in electric vehicle operation.

[0041] A data logger is connected to the temperature sensors T1, T2, and T3 to record thermal measurements during testing. The data logger captures temperature readings at predetermined intervals, enabling analysis of thermal behavior under different operational conditions. The hardware setup enables

comprehensive testing scenarios including no-load conditions where the battery pack is monitored without external current draw, and load conditions where the DC motor and loading rheostat create controlled power demands. This configuration allows for collection of battery performance data across a range of operational states, from idle conditions to high-power discharge scenarios that simulate acceleration and sustained driving conditions in electric vehicles.

[0042] FIG. 1 illustrates a Random Forest ensemble learning methodology according to various embodiments. The Random Forest architecture processes a new sample input through multiple parallel decision trees, where each decision tree performs independent analysis of the input data. The new sample, represented as battery sensor data including temperature, voltage, and current measurements, is simultaneously fed to each decision tree in the ensemble.

[0043] Each decision tree in a Random Forest ensemble performs binary splits at branching nodes, evaluating feature thresholds to determine the optimal path through the tree. These splits partition the input feature space based on battery sensor measurements, leading to terminal leaf nodes with final predictions. The Random Forest uses a bagging approach, training individual decision trees on bootstrap samples of battery sensor data, including temperature, voltage, and current measurements. This introduces diversity among the trees and reduces correlation between their predictions.

[0044] The results from multiple decision trees are aggregated through averaging, producing a final prediction with reduced variance and improved robustness and accuracy in detecting abnormal temperature conditions in battery systems. The battery monitoring system employs four machine learning models: Random Forest, XGBoost, LightGBM, and CatBoost. Each algorithm offers unique advantages for analyzing battery sensor data and detecting thermal anomalies.

[0045] The Random Forest achieves high accuracy and R values through its ensemble learning approach. XGBoost uses gradient boosting to correct errors iteratively, LightGBM optimizes computational efficiency with leaf-wise growth, and CatBoost handles mixed data types with ordered boosting. The ensemble learning architecture of Random Forest excels in identifying temperature

anomalies, indicating potential thermal runaway, capacity degradation, or other battery faults, enhancing safety in electric vehicle applications.

[0046] FIG. 2 illustrates an XGBoost ensemble machine learning architecture for battery fault detection according to various embodiments. The XGBoost framework processes training data through multiple parallel decision trees, where Decision Tree 1, Decision Tree 2, and Decision Tree N each analyze the battery sensor data independently. The training data comprises temperature, voltage, and current measurements collected from the battery monitoring system during various operational conditions.

[0047] The XGBoost ensemble algorithm uses multiple decision trees to improve prediction accuracy for battery fault detection and temperature anomaly identification. Each tree produces individual results, which are aggregated and averaged to form the final predictions. The algorithm employs a sequential boosting methodology, where each tree corrects errors from previous ones, progressively enhancing prediction accuracy by focusing on misclassified battery sensor measurements.

[0048] XGBoost uses gradient descent optimization to minimize prediction errors, adjusting model parameters systematically. The formula $\theta = \theta - \eta \nabla J(\theta)$ guides these updates, where θ represents model parameters, η is the learning rate, and $\nabla J(\theta)$ is the gradient of the loss function. Regularization techniques are incorporated to prevent overfitting, ensuring robust performance on new battery data. The regularized loss function $L(\theta) = \sum L(y_i, \hat{y}_i) + \Omega(\theta)$ penalizes model complexity, balancing prediction accuracy and generalization.

[0049] The iterative improvement process follows $F_m(x) = F_{m-1}(x) + \gamma h_m(x)$, where $F_m(x)$ is the prediction at iteration m , $F_{m-1}(x)$ is from the previous iteration, γ is the learning rate, and $h_m(x)$ is the new tree's prediction. This approach helps the algorithm build an ensemble that targets residual errors from previous iterations. XGBoost's architecture captures nonlinear relationships in battery sensor data, detecting complex fault patterns, temperature variations, and electrical changes, ensuring stable performance across various battery chemistries and operational conditions in electric vehicles.

[0050] FIG. 3 illustrates the leaf-wise tree growth methodology employed in the LightGBM algorithm according to various embodiments. The leaf-wise growth strategy demonstrates a sequential tree construction process across four distinct stages, where the algorithm systematically expands the decision tree by selecting the leaf node that yields maximum loss reduction at each iteration.

[0051] The initial tree structure has a root node connected to two child nodes. One child node, a filled solid black circle, is selected for further splitting, while the other, an open circle, remains a terminal leaf node. This selective splitting prioritizes nodes with the greatest information gain.

[0052] The previously selected filled node splits into two new child nodes, both filled circles, indicating their potential for future splitting. The tree now has three levels: the root node, an intermediate layer, and a bottom layer with two filled nodes and one unfilled node. This pattern focuses computational resources on promising nodes.

[0053] One of the filled nodes from the second stage splits based on the highest information gain, expanding into two open circles as new leaf nodes. The tree becomes more complex, demonstrating the asymmetric growth pattern of the leaf-wise approach.

[0054] Another leaf node undergoes selection and splitting. A previously open circle transforms into a filled circle, indicating selection for expansion, while new child nodes are added below. The tree exhibits an asymmetric structure, increasing depth selectively rather than uniformly.

[0055] The leaf-wise growth methodology prioritizes depth over breadth, enabling faster convergence compared to level-wise strategies. This selective expansion allows efficient navigation of complex feature spaces, focusing on regions with maximum discriminative power for battery fault detection.

[0056] LightGBM uses a histogram-based approach to group continuous feature values from battery sensor measurements into discrete bins, reducing memory consumption and computational complexity. This binning process facilitates rapid tree construction by evaluating only bin boundaries, significantly

reducing computational overhead, especially in large-scale battery monitoring datasets.

[0057] Combining leaf-wise growth with histogram-based feature processing, LightGBM achieves faster convergence and improved accuracy in analyzing high-dimensional battery sensor datasets. The algorithm identifies complex patterns in temperature, voltage, and current measurements, indicating fault conditions, thermal instability, or capacity degradation in lithium-ion batteries. Its computational efficiency makes it suitable for real-time battery monitoring.

[0058] However, the asymmetric tree growth can lead to overfitting in small or noisy datasets, necessitating careful hyperparameter tuning to balance model complexity and generalization capability. Parameters like maximum tree depth, minimum samples per leaf, regularization coefficients, learning rate, and the number of estimators need optimization for effective battery monitoring applications.

[0059] FIG. 4 illustrates a systematic fault detection and monitoring process flowchart according to various embodiments. The flowchart demonstrates a comprehensive workflow that begins with a start node and progresses through two distinct operational phases designed to ensure reliable battery fault detection and continuous system monitoring.

[0060] Phase 1 involves data processing operations foundational to the battery monitoring system. It starts with sensor data acquisition, where temperature sensors (T1, T2, T3) gather thermal data from the battery pack, alongside voltage and current readings from electrical circuits. This real-time data is captured at set intervals to create a continuous stream for analysis.

[0061] Next, data preprocessing cleans, filters, and standardizes the collected measurements. This stage involves noise removal, handling missing values, and normalizing sensor readings to ensure consistent data quality. A feedback loop allows for continuous refinement of data quality and preprocessing algorithms based on observed data characteristics and system performance.

[0062] Phase 2 focuses on model training for battery fault detection using machine learning algorithms. It includes selecting and training/testing models like

Random Forest, XGBoost, LightGBM, and CatBoost. These operations are iteratively refined based on performance evaluations. The preprocessed data is split into training and testing sets to validate and assess model performance.

5 **[0063]** A decision node labeled "Fault Detection" evaluates if abnormal conditions are present based on model predictions. It analyzes temperature, voltage, and current data to determine if faults exist that require immediate action. If no fault is detected, the system continues monitoring, maintaining a feedback loop for ongoing surveillance and returning to the sensor data acquisition stage for uninterrupted operations.

10 **[0064]** If a fault is detected, the system triggers an alert, initiating response protocols to address the issue. This alert mechanism ensures timely intervention to prevent potential hazards or failures.

15 **[0065]** The battery monitoring system operates in two modes: load and no-load conditions. Load conditions simulate active battery usage with current draws up to 4A, replicating electric vehicle demands like acceleration and high-power accessory usage. This testing uses a DC motor and loading rheostat to create controlled scenarios, revealing thermal behavior under realistic demands.

20 **[0066]** No-load conditions simulate idle states with minimal current draw, such as vehicle parking or overnight storage. This mode detects faults that may develop during reduced electrical activity, including self-discharge anomalies and internal resistance changes.

25 **[0067]** Comprehensive monitoring across both modes enables detection of fault patterns under different scenarios. Thermal stress patterns appear during high-current loads, indicating potential fault conditions or capacity degradation. Voltage drop patterns during load testing signal deteriorating battery health. Capacity degradation is evident through reduced voltage maintenance and altered thermal behavior during charge/discharge cycles.

30 **[0068]** The fault detection system uses dynamic threshold detection, adjusting temperature thresholds based on operational conditions to prevent false alarms while maintaining sensitivity to genuine faults. Critical temperature thresholds range from 46.2°C to 50.2°C, calibrated based on the thermal characteristics of

DMEC INR18650-26E lithium-ion cells and specific sensor locations. This ensures accurate fault detection across all operational conditions.

[0069] FIG. 5 illustrates a comprehensive machine learning pipeline for battery fault detection and analysis according to various embodiments. The pipeline workflow commences with library importation and Google Drive mounting operations that establish the computational environment and data access infrastructure for battery sensor data processing. The initial stages include importing machine learning libraries, data processing modules, and visualization tools required for comprehensive battery fault analysis.

[0070] The workflow proceeds through dataset listing operations that identify available battery sensor datasets stored in the cloud-based data repository. The dataset listing functionality enables selection of specific battery monitoring datasets corresponding to different operational conditions, battery configurations, or testing scenarios. The dataset selection process provides flexibility to analyze battery data from various sources including laboratory testing, field measurements, or simulated operational conditions.

[0071] The data preprocessing stage implements comprehensive data cleaning and preparation operations to ensure high-quality inputs for machine learning model training. The preprocessing operations address common data quality issues encountered in battery sensor measurements including missing values, measurement noise, and inconsistent sampling intervals that can affect model performance and prediction accuracy.

[0072] Missing value handling techniques are implemented through multiple approaches tailored to different data characteristics and missing value patterns.

Linear interpolation addresses isolated missing values using the formula $x_i^* = \frac{x_{i-1} + x_{i+1}}{2}$, where x_i represents the interpolated value, x_{i-1} denotes the preceding measurement, and x_{i+1} represents the subsequent measurement. Linear interpolation provides accurate estimates for missing temperature, voltage, or current measurements when the missing values occur sporadically within continuous sensor data streams.

[0073] Median imputation addresses missing values in skewed distributions commonly encountered in battery sensor data where extreme values or outliers may distort mean-based imputation approaches. Median imputation replaces missing values with the median value of the corresponding feature, providing robust estimates that are less sensitive to outliers or extreme measurements that may occur during fault conditions or measurement errors.

[0074] Iterative imputation handles complex feature dependencies where missing values in one sensor measurement may correlate with values in other sensor channels. The iterative imputation process models each feature with missing values as a function of other features, enabling sophisticated missing value estimation that accounts for inter-sensor relationships and temporal dependencies in battery monitoring data.

[0075] The feature engineering process creates enhanced representations of the battery sensor data through the generation of lag features, moving averages, and exponential moving averages that capture temporal patterns and trends in battery behavior. Lag features are constructed using the formula $Lag_k = x_{t-k}$, where Lag_k represents the lag feature at time step k, and x_{t-k} denotes the sensor measurement from k time steps in the past. Lag features enable the machine learning models to incorporate historical sensor values when making predictions about current battery conditions.

[0076] Moving averages are calculated using the formula $MA_k = \frac{1}{k} \sum_{i=1}^k x_i$, where MA_k represents the moving average over k time steps, and x_i denotes individual sensor measurements within the averaging window. Moving averages smooth short-term fluctuations in battery sensor data while preserving longer-term trends that may indicate developing fault conditions or gradual performance degradation.

[0077] Exponential moving averages follow the recursive formula $EMA_t = \alpha x_t + (1-\alpha)EMA_{t-1}$, where EMA_t represents the exponential moving average at time t, α constitutes the smoothing factor, x_t denotes the current sensor measurement, and EMA_{t-1} represents the previous exponential moving average value. The smoothing factor α controls the weighting between recent and historical

measurements, with higher values emphasizing recent data and lower values providing greater smoothing of temporal variations.

[0078] Feature engineering operations enhance machine learning models' ability to detect temporal patterns in battery behavior, indicating potential faults, capacity degradation, or thermal instabilities. Lag features detect delayed responses, while moving averages reveal gradual trends. Exponential moving averages adaptively smooth data, filtering measurement noise.

[0079] The train-test split operation divides the battery sensor dataset in an 80-20 ratio, maintaining temporal continuity through time-series splitting. This ensures training data precedes test data, preventing data leakage and enabling realistic model performance evaluation on future conditions.

[0080] The 80-20 split allocates 80% of data for training and 20% for testing, providing sufficient training data and adequate test data for robust performance assessment. The training dataset helps models learn battery behavior patterns, while the test dataset evaluates model generalization to unseen conditions.

[0081] Feature standardization normalizes measurements using the formula $x'_i = \frac{x_i - \mu}{\sigma}$, where x'_i is the standardized value, x_i is the original measurement, μ is the mean, and σ is the standard deviation. This ensures all measurements contribute equally to model training, preventing dominance by features with larger numerical ranges. Standardization normalizes temperature, voltage, and current measurements, facilitating effective model training and stability.

[0082] Outlier detection and removal use the Interquartile Range (IQR) method to identify and eliminate anomalous measurements. The IQR bounds are calculated as $(Q1 - 1.5 \times IQR)$ for the lower bound and $(Q3 + 1.5 \times IQR)$ for the upper bound, where $Q1$ and $Q3$ are the first and third quartiles, respectively. This process removes extreme values, improving model robustness and fault detection accuracy.

[0083] Model training and evaluation use Random Forest, XGBoost, LightGBM, and CatBoost algorithms with preprocessed features. Evaluation metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), and R

Score to assess prediction accuracy and performance across different battery conditions.

[0084] The fault detection decision node evaluates model predictions to determine whether abnormal battery conditions exist based on temperature thresholds, voltage variations, or current anomalies detected by the machine learning algorithms. The decision process triggers appropriate response actions including continued monitoring for normal conditions or alert generation for detected fault conditions.

[0085] The result visualization and model identification stages generate graphical representations of model performance and identify the algorithm that achieves the highest accuracy for the specific battery monitoring application. The final prediction and fault data storage operations preserve analysis results in structured formats that enable further analysis, reporting, and integration with battery management systems.

[0086] FIG. 6 illustrates battery temperature fault detection with difference analysis comparing normal and abnormal operational conditions according to various embodiments. The graph displays temperature variations over a time period spanning approximately 35 seconds, with the horizontal axis representing time in seconds and the vertical axis showing temperature measurements in degrees Celsius ranging from approximately 25°C to 60°C.

[0087] The normal condition curve appears as a blue line that exhibits a gradual temperature increase from approximately 25°C at the initial time point to approximately 30°C over the 35-second monitoring period. The normal condition curve demonstrates smooth, controlled thermal behavior characteristic of stable battery operation without fault conditions. The gradual temperature rise reflects normal thermal response during standard battery operation where internal resistance and ambient conditions contribute to modest temperature increases within acceptable operational parameters.

[0088] The abnormal condition curve appears as a brown line that demonstrates a steep temperature escalation from approximately 25°C at the initial measurement point to nearly 50°C by the 20-second mark. The abnormal condition

curve exhibits rapid thermal acceleration that indicates developing fault conditions or thermal instability within the battery system. Following the steep temperature rise, the abnormal condition curve plateaus near 50°C, suggesting that the fault condition reaches a steady state or that thermal protection mechanisms limit further temperature increases.

[0089] The fault classification system categorizes battery thermal conditions into three distinct operational ranges based on temperature measurements. Normal conditions encompass temperature ranges from 25.5°C to 35°C, representing safe operational parameters where the battery system functions within design specifications without thermal stress or fault indicators. Abnormal conditions span temperature ranges from 35°C to 48.9°C, indicating elevated thermal conditions that may precede fault development or suggest developing thermal instability requiring monitoring and potential intervention.

[0090] Blowout conditions encompass temperature ranges from 48.9°C to 60°C, representing dangerous thermal conditions that pose safety risks and indicate severe fault conditions requiring immediate corrective action. The blowout temperature range corresponds to thermal conditions that may lead to thermal runaway, capacity degradation, or safety hazards in lithium-ion battery systems.

[0091] The fault classification system employs corresponding color coding to provide visual identification of different thermal conditions. Normal temperature conditions are represented by blue coloring, providing clear visual indication of safe operational states. Abnormal temperature conditions are indicated by yellow coloring, signaling elevated thermal conditions that require attention. Blowout conditions are marked by red coloring, providing immediate visual warning of dangerous thermal states that demand urgent intervention.

[0092] Detected fault markers appear as red X symbols scattered along the abnormal condition curve, identifying specific instances where temperature deviations exceed predefined thresholds established for fault detection. The red X symbols highlight measurement points where the machine learning algorithms detect anomalous thermal behavior based on temperature rate of change, absolute temperature values, or deviation from expected thermal patterns. The fault markers

enable precise identification of the temporal locations where abnormal thermal events occur within the monitoring period.

[0093] Temporal fault boundaries are marked by vertical dashed red lines positioned at approximately 6 seconds and 20 seconds, designated as Fault Start and Fault End respectively. The Fault Start marker at 6 seconds identifies the temporal point where abnormal thermal behavior begins, indicating the onset of conditions that deviate from normal battery thermal response. The Fault End marker at 20 seconds delineates the temporal boundary where the abnormal thermal event concludes or stabilizes, providing clear temporal isolation of the fault condition.

[0094] The temporal fault boundary identification demonstrates the system's capability to precisely isolate abnormal thermal events within the continuous monitoring data stream. The 14-second duration between Fault Start and Fault End represents the temporal extent of the abnormal thermal event, enabling quantitative analysis of fault development patterns and thermal response characteristics during fault conditions.

[0095] The battery monitoring system achieves fault detection accuracy exceeding 96% across multiple datasets encompassing various operational conditions and battery configurations. The high detection accuracy results from the combination of machine learning algorithms, dynamic threshold detection, and comprehensive sensor data analysis that enables reliable identification of abnormal thermal patterns while minimizing false positive detections.

[0096] The detection accuracy performance is validated across datasets representing different operational scenarios including load conditions with current draw up to 4A and no-load conditions simulating idle battery states. The consistent accuracy performance across diverse operational conditions demonstrates the robustness of the fault detection algorithms and their suitability for real-world battery monitoring applications in electric vehicle systems.

[0097] The fault detection system processes temperature measurements from the T1, T2, and T3 sensor locations to identify localized thermal anomalies and thermal gradient patterns that may indicate developing fault conditions. The multi-sensor approach enables detection of thermal variations across the battery pack that

might be missed by single-point temperature monitoring, providing comprehensive thermal surveillance of the battery system.

[0098] The difference analysis methodology compares temperature measurements between normal and abnormal operational states to identify characteristic patterns that distinguish fault conditions from normal thermal variations. The analysis reveals that normal conditions exhibit temperature variations within $\pm 2.5^{\circ}\text{C}$, while abnormal conditions demonstrate deviations exceeding $\pm 6.8^{\circ}\text{C}$, providing quantitative criteria for fault classification and threshold establishment.

10 [0099] The machine learning algorithms process the temperature difference patterns to establish dynamic fault thresholds that adapt to operational conditions while maintaining sensitivity to genuine fault conditions. The dynamic threshold approach prevents false alarms during normal temperature fluctuations associated with varying load conditions while ensuring reliable detection of abnormal thermal events that indicate developing battery faults or safety hazards.

[0100] FIG. 7 illustrates a comparative line graph of temperature behavior under normal and abnormal battery operating conditions according to various embodiments. The graph demonstrates distinct thermal response patterns that enable differentiation between safe operational states and potentially hazardous fault conditions through analysis of temperature trajectory characteristics and thermal rate of change parameters.

[0101] The normal condition curve shows a controlled temperature rise from 25°C to 35°C over 20 seconds, indicating stable battery operation with balanced heat generation and dissipation. This stabilization reflects effective thermal management within acceptable operational parameters.

[0102] In contrast, the abnormal condition curve exhibits a rapid temperature increase from 30°C to nearly 50°C within the same timeframe, suggesting fault conditions where heat generation surpasses dissipation capacity, potentially leading to thermal instability or runaway. The steep temperature gradient signals imminent dangerous states requiring immediate intervention.

[0103] Blowout markers, clustered in the higher temperature region of the abnormal curve, indicate critical thermal events where temperatures exceed safe operational thresholds for lithium-ion batteries. These markers help fault detection algorithms identify dangerous states, as they correspond to temperatures approaching thermal runaway initiation.

[0104] Experimental data analysis shows normal operations have temperature deviations within 25°C, reflecting stable thermal management and consistent electrochemical behavior. This narrow deviation band allows for tight tolerance limits for detecting normal operations while maintaining sensitivity to thermal anomalies.

[0105] Abnormal states show temperature deviations up to 68°C, nearly three times that of normal operations, indicating thermal instability and potential faults. This expanded deviation range provides a basis for threshold establishment and fault classification.

[0106] Operational time analysis across datasets reveals 32% to 41% of time classified as abnormal based on temperature deviations, underscoring the general reliability of lithium-ion batteries while highlighting the need for continuous monitoring to detect faults. Variations in abnormal time percentages reflect differences in operational conditions, battery age, or environmental factors.

[0107] Comparative visualization aids fault detection algorithms by analyzing temperature trajectories and rates of change. This analysis distinguishes normal gradual heating from rapid abnormal escalation, identifying developing faults before absolute temperature thresholds are exceeded.

[0108] The fault detection algorithms process the comparative thermal patterns to establish dynamic classification boundaries that separate safe operational states from dangerous thermal anomalies. The classification boundaries account for operational context including load conditions, ambient temperature, and battery state of charge to prevent false classifications while maintaining sensitivity to genuine fault conditions. The dynamic boundary adjustment enables robust fault detection across varying operational scenarios encountered in electric vehicle applications.

[0109] The temperature trajectory analysis supports predictive maintenance strategies by enabling early identification of thermal behavior changes that precede complete fault development. The predictive capability allows maintenance interventions before critical failures occur, reducing safety risks and extending battery system operational life. The trajectory-based prediction approach analyzes thermal response patterns to forecast fault development timelines and recommend appropriate maintenance actions based on observed thermal behavior trends.

[0110] The comparative analysis methodology provides quantitative metrics for battery health assessment through thermal behavior characterization under different operational conditions. The health assessment metrics derived from temperature trajectory analysis enable trending of battery thermal performance over time, identification of gradual degradation patterns, and prediction of remaining useful life based on thermal response characteristics. The thermal health metrics complement traditional electrical performance indicators to provide comprehensive battery condition monitoring capabilities.

[0111] FIG. 8 illustrates a scatter plot comparing actual versus predicted temperature values under normal operating conditions for the T1 dataset according to various embodiments. The scatter plot displays temperature measurements over a 30-second monitoring period, with the horizontal axis representing time in seconds ranging from 0 to approximately 30 seconds, and the vertical axis showing temperature values spanning from approximately 26°C to 32°C.

[0112] Temperature measurements, represented by blue circular markers, follow an S-shaped curve from 26°C to 32°C over 30 seconds, reflecting typical thermal responses in lithium-ion batteries. The Random Forest regression model's predictions, shown as red X markers, closely align with these measurements, indicating high predictive accuracy. Performance metrics validate the model's effectiveness: an R value of 0.9997 shows it explains 99.97% of temperature variance, a Mean Squared Error of 0.00036 indicates minimal prediction error, and a Mean Absolute Error of 0.00376 confirms precise thermal monitoring.

[0113] FIG. 9 illustrates a scatter plot comparing actual versus predicted battery temperature values under abnormal operating conditions according to

various embodiments. The scatter plot displays the relationship between actual temperature measurements and Random Forest model predictions across a temperature range spanning from approximately 30°C to 50°C, representing the elevated thermal conditions characteristic of abnormal battery operational states.

5 **[0114]** Abnormal temperature measurements, represented by orange circular markers, range from 30°C to 50°C and are collected during fault conditions, thermal stress events, or other abnormal scenarios. These markers reflect the thermal characteristics observed during various abnormal battery conditions.

10 **[0115]** The Random Forest model's predictions are shown as blue X-shaped markers, which exhibit a strong linear correlation with the actual temperature measurements. The alignment of blue X markers with orange circular markers along a diagonal trend line indicates the model's maintained predictive accuracy under abnormal thermal conditions. This correlation demonstrates the model's ability to capture the relationship between input features and temperature outcomes
15 across abnormal operational conditions.

20 **[0116]** The strong linear correlation between actual and predicted values under abnormal conditions validates the Random Forest algorithm's robustness. The alignment of prediction markers with actual measurements along the diagonal trend line confirms the model's predictive capability during fault conditions, thermal stress events, and other challenging scenarios. The consistent correlation pattern across the temperature range indicates model stability under varying degrees of thermal anomaly severity.

25 **[0117]** Performance metrics under abnormal conditions show continued model effectiveness despite the challenging environment. The coefficient of determination (R^2) achieves a value of 0.9989, indicating that the model explains 99.89% of the variance in temperature measurements during abnormal conditions. The high R^2 value confirms the model's predictive accuracy even when processing thermal data from fault scenarios or thermal stress events.

30 **[0118]** The Mean Squared Error (MSE) under abnormal conditions increases to 0.0156, reflecting the additional prediction challenge posed by abnormal thermal behavior. The Mean Absolute Error (MAE) reaches 0.0870, representing acceptable

prediction accuracy for fault detection and thermal monitoring applications during challenging scenarios. The evaluation metrics framework uses standardized mathematical formulations to ensure consistent and objective model performance assessment across different conditions and datasets. The MAE calculation provides
5 a direct measure of average prediction error magnitude, enabling straightforward interpretation of model accuracy.

[0119] The Mean Squared Error calculation employs the formula $MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$, where the squared difference between actual and predicted values emphasizes larger prediction errors while providing a differentiable loss
10 function suitable for model optimization. The MSE metric penalizes large prediction errors more heavily than small errors, making the metric particularly sensitive to outliers or significant prediction failures that could compromise battery monitoring effectiveness.

[0120] The coefficient of determination follows the mathematical relationship
15 $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{2 \sum_{i=1}^n (y_i - \bar{y})^2}$, where $\bar{y}$ represents the mean of actual temperature measurements. The R^2 metric quantifies the proportion of variance in temperature measurements that the model successfully explains, with values approaching 1.0 indicating superior predictive performance and values near 0.0 suggesting poor model effectiveness.

20 **[0121]** The classification performance metrics complement the regression evaluation framework to provide comprehensive assessment of the battery monitoring system's fault detection capabilities. The classification accuracy exceeds 98.12%, indicating that the system correctly identifies normal and abnormal battery conditions in more than 98% of evaluated cases. The high
25 accuracy performance demonstrates the system's reliability for distinguishing safe operational states from potentially hazardous fault conditions across diverse battery operational scenarios.

[0122] The precision metric achieves 97.03%, representing the proportion of predicted abnormal conditions that correspond to actual abnormal battery states.

The high precision value indicates that the system minimizes false positive detections, reducing unnecessary alerts or interventions that could disrupt normal battery operations. The precision performance ensures that fault detection alerts correspond to genuine abnormal conditions requiring attention or corrective action.

5 **[0123]** The recall metric reaches 96.23%, quantifying the proportion of actual abnormal conditions that the system successfully identifies through its fault detection algorithms. The high recall value demonstrates the system's sensitivity to genuine fault conditions, ensuring that dangerous thermal events or developing battery problems receive appropriate detection and response. The recall
10 performance minimizes false negative detections that could allow hazardous conditions to persist without detection.

[0124] The F1-score achieves 97.57%, representing the harmonic mean of precision and recall metrics that provides a balanced assessment of classification performance. The high F1-score indicates that the system maintains excellent
15 performance across both precision and recall dimensions, achieving optimal balance between false positive minimization and false negative prevention. The F1-score performance validates the system's suitability for safety-critical battery monitoring applications where both detection reliability and alert accuracy are paramount.

20 **[0125]** The comprehensive performance metrics demonstrate the battery monitoring system's robust predictive capabilities across varying operational conditions including normal thermal states and abnormal fault scenarios. The consistent high performance across multiple evaluation metrics validates the machine learning approach for battery thermal monitoring and fault detection
25 applications. The minimal prediction error characteristics enable precise temperature forecasting that supports predictive maintenance strategies and proactive fault prevention in electric vehicle battery systems.

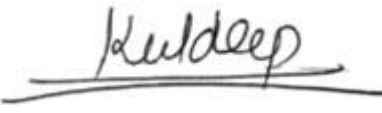
[0126] The performance evaluation framework provides quantitative validation of the system's effectiveness for real-world battery monitoring
30 applications where accurate thermal prediction and reliable fault detection are essential for safety and operational efficiency. The combination of regression and

classification metrics ensures comprehensive assessment of both temperature prediction accuracy and fault detection reliability across the full spectrum of battery operational conditions encountered in electric vehicle applications.

5 [0127] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This 07th day of November 2025

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WE CLAIM:

1. A battery temperature fault detection system for electric vehicles, comprising:

5 a plurality of temperature sensors configured to monitor thermal conditions of a lithium-ion battery pack during operation;

 a data processing module configured to analyze sensor data from the temperature sensors using machine learning analysis to identify thermal patterns;

 a fault detection module configured to identify abnormal temperature
10 conditions based on the machine learning analysis by comparing predicted temperature values against actual sensor readings; and

 an alert generation system configured to provide output signals when the abnormal temperature conditions are detected.

2. The system of claim 1, wherein the plurality of temperature sensors
15 comprises at least three temperature sensors positioned at strategic locations on the lithium-ion battery pack to capture thermal distribution patterns and detect localized heating events.

3. The system of claim 1 or 2, wherein the fault detection module is configured to implement dynamic threshold adjustment that adapts detection
20 sensitivity based on operational conditions including load states and battery aging characteristics.

4. The system of any of claims 1 to 3, wherein the machine learning analysis achieves classification accuracy exceeding 96% for fault detection and regression
25 performance with R-squared values above 0.99 for temperature prediction under both normal and abnormal operating conditions.

5. The system of any of claims 1 to 4, wherein the alert generation system is configured to provide graduated alert levels including early warning notifications for minor temperature deviations and critical alerts for conditions approaching thermal runaway thresholds.

30 6. A method for detecting abnormal battery temperature conditions in electric vehicles, comprising:

collecting temperature sensor data from a lithium-ion battery pack using a plurality of temperature sensors;

processing the temperature sensor data using machine learning analysis to identify thermal patterns and predict temperature values;

5 detecting abnormal temperature conditions based on the machine learning analysis by comparing the predicted temperature values against actual sensor measurements; and

generating alerts when the abnormal temperature conditions are detected that exceed predetermined safety thresholds.

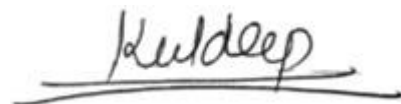
10 7. The method of claim 6, further comprising preprocessing the temperature sensor data by performing outlier removal using interquartile range methods, normalization of sensor measurements, and feature engineering to enhance the machine learning analysis.

8. The method of claim 6 or 7, wherein the detecting step comprises real-time monitoring including continuous analysis of temperature trends, rate of temperature change, and correlation patterns between temperature measurements from the plurality of temperature sensors.

9. The method of any of claims 6 to 8, wherein the processing step comprises implementing multiple ensemble learning techniques including Random Forest, XGBoost, LightGBM, and CatBoost to analyze the temperature sensor data and selecting an optimal technique based on performance evaluation.

10. The method of any of claims 6 to 9, wherein the detecting step comprises adaptive threshold detection that adjusts threshold values based on battery load conditions, ambient temperature, battery age, and historical performance patterns.

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ABSTRACT

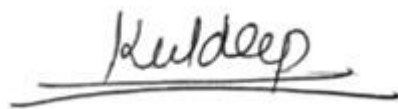
MACHINE LEARNING-BASED BATTERY TEMPERATURE FAULT DETECTION SYSTEM FOR ELECTRIC VEHICLES

The present disclosure provides a battery temperature fault detection system for electric vehicles that includes a plurality of temperature sensors (T1, T2, T3) configured to monitor thermal conditions of a lithium-ion battery pack during operation, a data processing module configured to analyze sensor data from the temperature sensors using machine learning analysis to identify thermal patterns, a fault detection module configured to identify abnormal temperature conditions based on the machine learning analysis by comparing predicted temperature values against actual sensor readings, and an alert generation system configured to provide output signals when the abnormal temperature conditions are detected. The system employs ensemble learning algorithms including Random Forest, XGBoost, LightGBM, and CatBoost to process temperature, voltage, and current measurements from the battery pack, enabling detection of thermal anomalies before thermal runaway conditions develop.

(Fig. 1)

[To be published along with figure 1]

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